

Title: Nonstandard Probabilities in an Internal Probability Space

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Standard probability theory assigns probability zero to any individual outcome of a continuous random variable, even when that outcome is entirely possible. This thesis argues that this result, while mathematically consistent, is conceptually unsatisfying, and proposes an alternative framework using hyperreal numbers.

The construction follows the ultrapower approach to the hyperreal number system ${}^*\mathbb{R}$, based on a fixed nonprincipal ultrafilter on $\mathbb{N}$, as developed by Goldblatt (1998). The thesis builds the necessary measure-theoretic foundations, including σ -algebras, Lebesgue measure, internal and hyperfinite sets, and the Loeb measure, before defining an internal probability space over a hyperfinite sample space.

The central result shows that, under this framework, the nonstandard probability of a uniform hypercontinuous random variable X taking a specific value a is infinitesimal but strictly positive:

$${}^*\mathbb{P}(X = a) \simeq 0, \quad \text{yet} \quad {}^*\mathbb{P}(X = a) > 0$$

The thesis also establishes the equivalence between the Lebesgue measure and the Loeb measure constructed on an internal hyperfinite partition of the real line.